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Evaluation of the relationship between vitamin levels and symptom severity in adults with attention-deficit/hyperactivity disorder

Demiryürek Esra[1]✉ & Kocayiğit Havva[2]

This cross-sectional study aimed to compare serum vitamin B12, 25-hydroxyvitamin D (25-OHD), ferritin, and iron levels between adults diagnosed with attention deficit hyperactivity disorder (ADHD) and healthy controls, and to evaluate the relationship between these biochemical markers and the severity of ADHD symptoms. In this cross-sectional study, a total of 35 adults with ADHD and 36 age- and sex-matched healthy controls who presented to a psychiatry outpatient clinic in Sakarya in 2024 were included. Participants' demographic data and clinical characteristics were collected through interviews, while laboratory findings were examined using blood samples taken at the time of inclusion in the study. The severity of ADHD symptoms was assessed using the Adult ADHD Self-Report Scale (ASRS) and the Wender–Utah Rating Scale (WURS). Serum vitamin B12 and 25-hydroxyvitamin D levels were significantly lower in the ADHD group compared with controls ($p < 0.001$). No significant differences were observed in ferritin and iron levels ($p > 0.05$). Vitamin B12 and 25-hydroxyvitamin D levels showed moderate-to-strong negative correlations with symptom severity. For example, vitamin B12 levels correlated with ASRS inattention ($r = -0.54$, $p = 0.00009$) and WURS inattention ($r = -0.56$, $p = 0.000002$), while 25-hydroxyvitamin D levels correlated with ASRS hyperactivity/impulsivity ($r = -0.53$) and WURS depression ($r = -0.54$). These findings indicate that lower serum vitamin B12 and 25-hydroxyvitamin D levels are associated with greater ADHD symptom severity in adults. The results underscore the potential relevance of biochemical factors in adult ADHD and suggest that assessment of vitamin status may be considered in future clinical and research contexts.

Keywords ADHD, vitamin B12, 25-hydroxyvitamin D, ferritin, iron

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental condition that begins in childhood and may persist into adulthood. Adult ADHD can lead to a marked reduction in quality of life due to symptoms of inattention, impulsivity, and hyperactivity. The prevalence of adult ADHD in the general population has been reported to range between 2 and 5%. While neurotransmitter systems such as dopamine and norepinephrine are central to its pathophysiology, nutritional and biochemical factors involved in the synthesis and metabolism of these neurotransmitters may also play a contributory role[1,2].

Recent studies have demonstrated that vitamin D, vitamin B12, ferritin, and iron levels play important roles in neurodevelopmental processes and cognitive functioning. Accordingly, deficiencies in these micronutrients have been proposed as potential contributors to ADHD pathogenesis. Meta-analyses conducted in child and adolescent populations have shown significantly lower serum 25-hydroxyvitamin D levels in patients with ADHD compared with healthy controls[3]. Randomized controlled trials in children have also reported that vitamin D supplementation may lead to partial improvement in attention and hyperactivity symptoms[4,5].

Similarly, there is robust evidence indicating reduced ferritin levels in individuals with ADHD. A meta-analysis found that ferritin levels were significantly lower in pediatric ADHD patients compared with controls (SMD = − 0.40; 95% CI = − 0.66 to − 0.14), whereas findings for serum iron levels have been inconsistent[4,5].

[1]Private Clinic, Sakarya, Türkiye, Turkey. [2]Sakarya University, Sakarya, Türkiye, Turkey. ✉email: esraozdil09@gmail.com

Studies on vitamin B12 have likewise suggested that B12 levels may be lower in children and adolescents with ADHD[6,7].

Despite robust associations reported in pediatric populations, these findings cannot be directly extrapolated to adult ADHD. The clinical presentation and underlying neurobiological mechanisms of ADHD evolve substantially across the lifespan, and evidence derived from children and adolescents may not fully capture the pathophysiology of the disorder in adulthood. Accordingly, the number of large-sample or prospective studies examining micronutrient-related parameters specifically in adult ADHD populations remains limited.

In addition to core dopaminergic and noradrenergic dysfunction, accumulating evidence suggests that micronutrients involved in neurotransmitter synthesis, neurodevelopment, and neurometabolic regulation may contribute to ADHD symptom expression. Vitamin B12 plays a critical role in one-carbon metabolism and methylation pathways essential for neuronal integrity, monoaminergic neurotransmission, and cognitive functioning; deficiency has been associated with attentional impairment, cognitive slowing, and neuropsychiatric symptoms[8,9]. Vitamin D is increasingly recognized as a neuroactive steroid that modulates dopaminergic signaling, neuroimmune processes, and neurotrophic factors, with experimental and clinical evidence suggesting its involvement in executive functioning and attentional regulation[10,11]. Iron-related parameters, including ferritin and serum iron, are of particular relevance given iron's role as a cofactor in dopamine synthesis, dopamine transporter regulation, and myelination. Importantly, peripheral iron indices reflect different aspects of iron metabolism and neurobiological functioning, warranting the assessment of both ferritin (as an index of iron stores) and serum iron[12,13].

However, findings regarding iron-related biomarkers in ADHD have been inconsistent across the literature. Meta-analytic evidence, largely derived from pediatric populations, has reported lower ferritin levels in some individuals with ADHD, while results for serum iron have been variable[14]. In adult populations, the relationship appears to be more complex and may depend on the clinical and hematological characteristics of the sample. For example, Demirci et al. (2017) reported higher ADHD symptom scores and an increased prevalence of ADHD diagnoses among adult women with iron deficiency anemia compared to healthy controls, suggesting that iron-related alterations may be more evident in the context of clinically manifest deficiency[15].

In contrast, findings in non-anemic adult ADHD samples have been less consistent. Lopez et al. (2019) reported that while iron deficiency may be observed in certain adult ADHD subgroups, such as individuals with comorbid restless legs syndrome, associations between ADHD and peripheral iron parameters are not consistently observed across broader non-anemic adult ADHD populations[16]. Consistent with these observations, the present study—which excluded individuals with a history of anemia—did not identify significant differences in ferritin or serum iron levels between adults with ADHD and healthy controls. Taken together, these findings support the assessment of both ferritin and serum iron to provide a more comprehensive evaluation of iron-related biomarkers, while suggesting that in the absence of overt anemia, peripheral iron indices may not show a robust association with ADHD symptom severity in adults.

In line with integrative perspectives highlighted in the literature, including work by Rucklidge and colleagues, multiple micronutrients may interact to influence brain function and behavioral regulation in ADHD, supporting the investigation of combined micronutrient profiles rather than isolated biomarkers[17]. Collectively, these neurobiological and biochemical considerations provide a strong rationale for examining serum vitamin B12, 25-hydroxyvitamin D, ferritin, and iron levels in relation to symptom severity in adult ADHD.

Previous research examining serum vitamin and iron parameters in adult ADHD populations remains underexplored and methodologically heterogeneous. A small number of studies have reported associations between adult ADHD and altered peripheral biomarkers such as ferritin, homocysteine, and oxidative stress parameters; however, these investigations often focused on selected biochemical markers, included heterogeneous clinical samples, or lacked well-matched healthy control groups[18,19]. Existing adult ADHD studies have primarily examined ferritin levels or oxidative stress markers, with variable consideration of symptom severity and limited control for confounding factors such as anemia, comorbid medical conditions, or psychotropic medication use[18,19]. Moreover, few studies have simultaneously assessed multiple micronutrients—particularly vitamin B12 and 25-hydroxyvitamin D—using standardized laboratory methods while directly correlating biochemical findings with validated ADHD symptom scales. In contrast, the present study employed strict inclusion and exclusion criteria, excluded individuals with anemia, systemic illness, or psychotropic medication use, and utilized age- and sex-matched healthy controls, enabling a more controlled evaluation of serum vitamin B12, 25-hydroxyvitamin D, ferritin, and iron levels in adults with ADHD. Additionally, by assessing symptom severity using both the ASRS and WURS and obtaining biochemical measurements on the same day as clinical evaluation, this study aimed to provide a more comprehensive and methodologically rigorous examination of the relationship between micronutrient status and symptom severity in adult ADHD.

Although ADHD is classified as a neurodevelopmental disorder with onset in childhood, its clinical presentation and underlying etiological mechanisms may differ between pediatric and adult populations. While childhood ADHD is predominantly driven by genetic vulnerability and early neurodevelopmental alterations, adult ADHD appears to be influenced by a broader range of factors, including age-related neurobiological changes and cumulative environmental, lifestyle, and biological influences across the lifespan[1,2]. These distinctions suggest that etiological factors identified in pediatric ADHD may not fully explain symptom persistence and expression in adulthood, highlighting the importance of investigating biochemical and nutritional correlates specifically in adult ADHD.

In light of these considerations, comparing serum vitamin B12, 25-hydroxyvitamin D, ferritin, and iron levels between adults with ADHD and healthy controls, as well as examining the relationship between these biomarkers and symptom severity, is clinically relevant and addresses an important gap in the current literature.

Materials and methods

This cross-sectional study aimed to compare serum vitamin B12, 25-hydroxyvitamin D, ferritin, and iron levels in adults diagnosed with attention-deficit/hyperactivity disorder (ADHD) with those of a healthy control group, and to evaluate the associations between these biochemical markers and the severity of ADHD symptoms.

Participants were recruited between January and December 2024 from a private psychiatry outpatient clinic in Sakarya, Türkiye. Thirty-five adults diagnosed with ADHD by a psychiatrist according to DSM-5 criteria and 36 age- and sex-matched healthy controls were included in the study.

Individuals aged 18–65 years with normal cognitive functioning and no history of anemia were eligible for inclusion. Exclusion criteria included comorbid neurological disorders, chronic systemic illnesses, current psychotropic medication use, pregnancy, obesity (to reduce potential metabolic confounding effects on vitamin D and related micronutrient levels), or active infection.

Demographic characteristics (age, sex, years of education), socioeconomic variables (marital status, living conditions, occupation), and detailed psychiatric history were obtained for all participants through clinical interviews. Psychiatric history assessment included information on childhood ADHD symptoms or diagnoses, history of other psychiatric disorders, past use of stimulant or other psychotropic medications, and family history of psychiatric conditions. Participants with current psychotropic medication use or comorbid psychiatric diagnoses were excluded from the study. Clinical reasons for referral and current psychiatric status were also evaluated. Participants reporting current use of vitamin B12, vitamin D, or iron supplements were excluded from the study. All participants were free from psychotropic medication at the time of blood sampling; prior stimulant exposure was assessed during clinical interviews but was not an exclusion criterion.

Symptom severity was assessed using the Wender–Utah Rating Scale (WURS) to evaluate retrospective childhood ADHD symptoms and the Adult ADHD Self-Report Scale (ASRS) to assess current symptom severity. The Adult ADHD Self-Report Scale (ASRS) and the Wender–Utah Rating Scale (WURS) are widely used standardized assessment instruments in ADHD research. Turkish validated versions of both scales were used in this study, and their use for research purposes does not require additional formal permission.

Venous blood samples were collected in the morning after an overnight fasting period on the same day as the clinical assessment to minimize potential diurnal and dietary effects on biochemical measurements. Serum vitamin B12, 25-hydroxyvitamin D, ferritin, and iron levels were analyzed using standardized laboratory procedures at the Sakarya University Training and Research Hospital Biochemistry Laboratory. Ferritin and vitamin B12 levels were assessed using chemiluminescent immunoassay on the ABBOTT ARCHITECT i2000SR analyzer; 25-hydroxyvitamin D levels were measured using the ChromSystems-based Architect 25-OH Vitamin D kit on the same platform, and serum iron levels were evaluated photometrically using the ABBOTT ARCHITECT c16000 device.

The study was conducted in accordance with the principles of the Declaration of Helsinki, and informed consent was obtained from all participants. The study protocol was approved by the Health Sciences Scientific Research Ethics Committee of Sakarya University School of Medicine (Approval No: 635; Document No: E-43012747-050.04-543041).

Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics for Windows, Version 27.0 (IBM Corp., Armonk, NY, USA; https://www.ibm.com/products/spss-statistics). The distribution of continuous variables was assessed using the Kolmogorov–Smirnov test. Normally distributed variables were expressed as mean ± standard deviation (SD), and categorical variables as frequencies and percentages. Between-group comparisons were performed using independent-samples t-tests for continuous variables and chi-square tests for categorical variables.

Although some variables approximated normal distribution, ASRS and WURS scores are derived from ordinal Likert-type items, and biochemical variables (particularly vitamin B12 and ferritin) showed skewness and outliers. Therefore, Spearman's rank correlation coefficient was preferred as a more robust non-parametric measure of association. A p-value < 0.05 was considered statistically significant.

Because multiple correlation analyses were conducted between biochemical markers and numerous symptom subscales, false discovery rate (FDR) correction using the Benjamini–Hochberg method was applied to reduce the risk of Type I error.

A post hoc power analysis was performed using G*Power software, Version 3.1.2 (Heinrich-Heine-University Düsseldorf, Germany; https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower). Based on the observed between-group differences in vitamin B12 and 25-hydroxyvitamin D levels, effect sizes were large (Cohen's $d > 0.8$), and the calculated statistical power exceeded 0.95, indicating adequate power for the primary biochemical comparisons. Post hoc power analysis for ASRS and WURS scale comparisons also demonstrated very large effect sizes (Cohen's d approximately 1.7 to > 6.0), with corresponding power values greater than 0.99. However, the study may have been underpowered to detect small effect sizes, particularly in correlation analyses and for biochemical parameters such as iron and ferritin.

Results

Comparison of demographic characteristics between the 35 adult ADHD patients and 36 healthy controls revealed no significant differences in age, sex, marital status, employment status, or years of education (all $p > 0.05$). Demographic characteristics of the participants are presented in Table 1.

All ASRS subscale and total scores were significantly higher in the ADHD group compared with controls ($p < 0.001$). Specifically, ADHD patients had significantly elevated ASRS inattention (31.37 ± 3.19), hyperactivity/impulsivity (24.57 ± 6.29), and total scores (55.94 ± 8.72) relative to controls (13.78 ± 3.37; 5.75 ± 3.67; 19.53 ± 6.20, respectively). WURS subscale and total scores, including irritability, depression, school problems,

Variable	ADHD Group ($n = 35$)	Control Group ($n = 36$)	p-value
Age	27.74 ± 7.25	26.72 ± 6.99	0.548
Female, n (%)	16 (45.7)	21 (58.3)	0.298
Married, n (%)	12 (34.3)	9 (25.0)	0.550
Single, n (%)	23 (65.7)	27 (75.0)	0.550
Employed, n (%)	16 (45.7)	17 (47.2)	0.604
Unemployed, n (%)	13 (37.1)	10 (27.8)	0.604
Years of Education	13.97 ± 2.47	14.06 ± 2.40	0.885

Table 1. Demographic Characteristics of the Patients. Note: Mean ± SD.

Scale Parameters		ADHD Group	Control Group	p-value
ASRS	Inattention	31.37 ± 3.19	13.78 ± 3.37	< 0.001
	Hyperactivity/Impulsivity	24.57 ± 6.29	5.75 ± 3.67	< 0.001
	Total Score	55.94 ± 8.72	19.53 ± 6.20	< 0.001
WURS	Irritability	16.57 ± 7.36	7.36 ± 0.64	< 0.001
	Depression	12.06 ± 3.05	5.36 ± 0.54	< 0.001
	School Problems	8.80 ± 3.58	3.08 ± 0.28	< 0.001
	Behavior/Impulsivity	14.06 ± 5.10	5.14 ± 0.35	< 0.001
	Inattention	19.89 ± 1.94	8.67 ± 1.49	< 0.001
	Total Score	71.40 ± 15.37	29.58 ± 2.25	< 0.001

Table 2. Comparison of ASRS and WURS Scores Between the ADHD and Control Groups. Note: ASRS: Adult ADHD Self-Report Scale; Mean ± SD. Note: WURS: Wender–Utah Rating Scale; Mean ± SD. No significant between-group differences were found in serum iron (67.17 ± 31.76 μg/dL vs. 69.38 ± 24.36 μg/dL; $p = 0.745$) or ferritin levels (41.34 ± 23.72 ng/mL vs. 41.71 ± 14.94 ng/mL; $p = 0.939$).

Parameter	ADHD Group	Control Group	p-value
Iron (μg/dL)	67.17 ± 31.76	69.38 ± 24.36	0.745
Ferritin (ng/mL)	41.34 ± 23.72	41.71 ± 14.94	0.939
Vitamin B12 (pg/mL)	194.83 ± 21.05	299.94 ± 126.99	< 0.001
25-OH Vitamin D (ng/mL)	22.73 ± 7.21	28.74 ± 5.67	< 0.001

Table 3. Comparison of Biochemical Parameters Between the ADHD and Control Groups. Note: Mean ± SD.

behavior/impulsivity, and inattention, were also significantly higher in the ADHD group (all $p < 0.001$). These findings confirm that both childhood-related symptoms and current attentional and behavioral impairments are substantially more pronounced in adults with ADHD than in controls. Detailed comparisons of ASRS and WURS scores between groups are presented in Table 2.

In contrast, vitamin B12 levels were significantly lower in the ADHD group (194.83 ± 21.05 pg/mL vs. 299.94 ± 126.99 pg/mL; $p < 0.001$). Similarly, 25-hydroxyvitamin D levels were markedly reduced in ADHD patients (22.73 ± 7.21 ng/mL vs. 28.74 ± 5.67 ng/mL; $p < 0.001$). These results indicate that serum vitamin B12 and 25-hydroxyvitamin D levels are significantly lower in adults with ADHD, whereas iron and ferritin levels do not differ significantly. Group comparisons of biochemical parameters are presented in Table 3.

Correlation analyses demonstrated significant negative associations between serum vitamin B12 and 25-hydroxyvitamin D levels and both ASRS and WURS subscale and total scores. Lower serum levels of these vitamins were associated with increased severity of inattention, hyperactivity/impulsivity, and behavioral symptoms. After applying false discovery rate (FDR) correction for multiple comparisons, the majority of these negative correlations remained statistically significant (e.g., ASRS inattention vs. vitamin B12: $r = -0.45$, adjusted $p = 0.00009$; WURS inattention vs. vitamin B12: $r = -0.53$, adjusted $p = 0.000002$), indicating robustness of the findings despite multiple testing.

Given the modest sample size and cross-sectional design, these associations should be interpreted as preliminary and hypothesis-generating. No significant correlations were found between iron or ferritin levels and any symptom scores (all $p > 0.05$). These findings suggest that lower serum micronutrient levels may be associated with ADHD symptom severity. Correlation analyses between micronutrient levels and ADHD symptom scores are summarized in Table 4.

	Biomarker	Scale	Spearman r	95% CI	Raw p-value	FDR-adjusted p
ASRS	B12	Inattention	−0.448	−0.622 to − 0.241	0.000091	**0.000163**
		Hyperactivity/Impulsivity	−0.375	−0.567 to − 0.143	0.001280	**0.001491**
		Total	−0.408	−0.589 to − 0.173	0.000412	**0.000571**
	25-OHD	Inattention	−0.467	−0.626 to − 0.269	0.000041	**0.000092**
		Hyperactivity/Impulsivity	−0.522	−0.666 to − 0.353	0.000003	**0.000028**
		Total	−0.495	−0.639 to − 0.321	0.000011	**0.000051**
WURS	B12	Irritability	−0.307	−0.504 to − 0.079	0.009316	**0.009316**
		Depression	−0.379	−0.585 to − 0.148	0.001120	**0.001440**
		School Problems	−0.374	−0.544 to − 0.146	0.001325	**0.001491**
		Behavior/Impulsivity	−0.329	−0.512 to − 0.133	0.005090	**0.005389**
		Inattention	−0.531	−0.695 to − 0.311	0.000002	**0.000028**
		Total	−0.469	−0.625 to − 0.276	0.000037	**0.000092**
	25-OHD	Irritability	−0.467	-0.637 to − 0.271	0.000030	**0.000089**
		Depression	−0.503	−0.656 to − 0.306	0.000008	**0.000047**
		School Problems	−0.409	−0.579 to − 0.197	0.000392	**0.000571**
		Behavior/Impulsivity	−0.488	−0.628 to − 0.317	0.000016	**0.000057**
		Inattention	−0.433	−0.639 to − 0.198	0.000161	**0.000264**
		Total	−0.461	−0.615 to − 0.283	0.000051	**0.000103**

Table 4. Spearman correlations between vitamin levels and ADHD symptom scores. Raw p-values and false discovery rate (FDR)–adjusted p-values (Benjamini–Hochberg procedure) are reported. ASRS: Adult ADHD Self-Report Scale; WURS: Wender–Utah Rating Scale.

Discussion

This study examined serum vitamin B12, 25-hydroxyvitamin D, ferritin, and iron levels in adults with ADHD compared with healthy controls and evaluated their associations with ADHD symptom severity. The main findings indicate that adults with ADHD had significantly lower vitamin B12 and vitamin D levels than controls, whereas ferritin and serum iron levels did not differ between groups. In addition, lower vitamin B12 and vitamin D levels were significantly associated with greater ADHD symptom severity, as assessed by both ASRS and WURS scores.

The observed reductions in vitamin D levels are broadly consistent with previous literature, which has predominantly focused on pediatric ADHD populations. Meta-analyses and randomized controlled trials in children and adolescents have reported lower serum 25-hydroxyvitamin D levels in individuals with ADHD and modest improvements in attention and hyperactivity symptoms following supplementation[3–5,20]. Rather than reiterating these well-established pediatric findings, the present results extend existing evidence by demonstrating that similar associations between vitamin D status and symptom severity are also evident in adult ADHD. Given vitamin D's involvement in dopaminergic signaling, neuroimmune regulation, and neurotrophic processes[10,11], low vitamin D levels may contribute to attentional and executive dysfunction across the lifespan.

Evidence regarding vitamin B12 in ADHD remains more limited. Nonetheless, low vitamin B12 status has been associated with impaired one-carbon metabolism, disrupted methylation pathways, alterations in monoaminergic neurotransmission, and reduced neurocognitive functioning[8,9]. Prior studies examining vitamin B12 in ADHD have largely focused on pediatric populations and have suggested lower B12 levels in affected individuals[7]. Consistent with these observations, our findings suggest that lower vitamin B12 levels may also be relevant to symptom expression in adults with ADHD, supporting the possibility that shared neurobiochemical mechanisms extend beyond childhood.

In contrast to findings reported in children, we did not observe significant differences in ferritin or serum iron levels between adults with ADHD and healthy controls. Meta-analytic evidence indicates reduced ferritin levels in pediatric ADHD samples, whereas findings for serum iron have been inconsistent[14,21]. Adult evidence appears more heterogeneous, and associations between iron-related parameters and ADHD may depend on clinical context. Prior studies have suggested that iron-related alterations are more evident in specific adult subgroups, such as individuals with iron deficiency anemia or comorbid conditions including restless legs syndrome, whereas findings in broader non-anemic adult ADHD samples are less consistent[15,16,18]. In the present study, exclusion of individuals with a history of anemia may have reduced variability in iron indices and limited the ability to detect group differences, potentially contributing to the null findings for ferritin and serum iron. These results suggest that peripheral iron parameters may be less informative in non-anemic adult ADHD populations.

This study contributes to the limited literature examining multiple micronutrients simultaneously in adults with ADHD. By focusing on an adult sample, employing strict inclusion and exclusion criteria, and using two validated symptom scales (ASRS and WURS), the findings highlight the potential relevance of vitamin B12 and vitamin D in adult ADHD symptomatology while suggesting a more limited role for ferritin and serum iron in non-anemic individuals. These results are consistent with integrative perspectives emphasizing the interaction of multiple micronutrients in influencing brain function and behavioral regulation in ADHD[6,17].

Several limitations should be considered when interpreting these findings. The relatively large variability observed in serum vitamin B12 levels, particularly in the control group, may reflect inter-individual differences in dietary intake, absorption, and subclinical nutritional status commonly reported in adult populations. The unusually high variability in control-group vitamin B12 levels (SD = 126.99 pg/mL, representing approximately 42% of the mean) also constitutes a methodological constraint that may limit the interpretability of our findings. Although statistically significant differences in serum vitamin B12 and 25-hydroxyvitamin D levels were observed, the present study did not aim to classify participants according to categorical deficiency or insufficiency thresholds; therefore, the clinical relevance of these differences should be interpreted with caution. The cross-sectional design precludes causal inference, and it remains unclear whether lower vitamin levels contribute to ADHD symptom severity or reflect behavioral and lifestyle factors associated with ADHD, such as dietary habits or reduced sunlight exposure. Multiple potential confounders—including dietary intake, vitamin supplementation, seasonal variation and sunlight exposure, physical activity, smoking status, and body composition—were not systematically assessed. Although obesity was excluded to reduce metabolic confounding, this may limit generalizability, as obesity is common in adult ADHD populations and influences vitamin D metabolism. In addition, exclusion of individuals with a history of anemia may have constrained the evaluation of iron-related biomarkers.

The relatively small sample size represents an additional limitation of this study. Although large effect sizes were observed for the between-group differences in serum vitamin B12 and 25-hydroxyvitamin D levels, the statistical power may have been insufficient to detect weaker associations, particularly in multiple correlation analyses. Therefore, very small p-values should not be interpreted as evidence of large clinical effects, but rather as statistical findings that require confirmation in larger and independent samples. Although false discovery rate (FDR) correction was applied, the correlation analyses should be interpreted cautiously due to multiple testing and their exploratory nature, and residual Type I error risk cannot be completely excluded. Finally, although prior psychotropic medication use was assessed, lifetime exposure and its potential long-term effects on nutritional status could not be fully quantified.

In conclusion, this study suggests that lower serum vitamin B12 and vitamin D levels are associated with greater symptom severity in adults with ADHD, whereas ferritin and serum iron levels may not differ in non-anemic adult populations. Assessment of vitamin status may be considered as part of a comprehensive evaluation in adults with ADHD; however, given the cross-sectional design, these findings should be interpreted cautiously. Longitudinal and interventional studies are needed to clarify causal relationships and to determine whether changes in micronutrient status are associated with clinical outcomes in adults with ADHD.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to ethical restrictions and patient confidentiality, but are available from the corresponding author on reasonable request.

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Author contributions

E.D. conceptualized and designed the study. E.D. and H.K. collected the data and conducted the analyses.H.K. performed the statistical analyses and prepared the figures and tables.E.D. supervised the study and interpreted the findings.E.D. and H.K. drafted the manuscript and critically revised it for important intellectual content.All authors approved the final version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to D.E.

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